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Department of Genetic Engineering and Biotechnology

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Case study report on Clubfoot disease

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Abstract:

Clubfoot is a congenital deformity that affects the feet, typically characterised by a medial deviation of the heel, a lateral deviation of the forefoot, and an inverted position of the ankle. In this case report, we present findings from a study of patients with clubfoot from various regions of Bangladesh. We collected information on demographics, clinical presentation, and treatment outcomes for patients diagnosed with clubfoot. Our study highlights the diversity in the presentation of clubfoot among patients in Bangladesh and the need for continued monitoring and treatment to achieve optimal outcomes. The results of this study will provide valuable insight into the epidemiology of clubfoot in Bangladesh and can inform future research and clinical practice.

Introduction:

Club foot, also known as congenital talipes equinovarus (CTEV), is a congenital deformity of the foot and ankle that affects 1 in 1,000 new-borns worldwide. The condition can be caused by a combination of genetics and environmental factors. Chromosomal abnormalities have been found in 30% of complex clubfoot cases and 2% of isolated clubfoot cases, including trisomy 18, 13, 21, sex chromosome abnormalities, micro-deletions and duplications. Symptoms of club foot include a twisted and inward facing top of the foot, an increased arch, and an inward facing heel. The affected leg or foot may also be slightly shorter and the calf muscles may be underdeveloped. Club foot can be treated through a variety of methods, including non-surgical methods such as casting and bracing, and surgical methods such as tenotomy and arthrodesis.

In Bangladesh, it is estimated that 5000 children are born with club foot each year, with an incidence rate of 1:900 births. Non-government organization, Walk for Life (WFL) has provided free treatment for over 17,500 Bangladeshi children with club feet since 2009, with local ownership and international support. Despite advancements in treatment, club foot continues to pose a significant burden to affected individuals and their families, both

physically and emotionally. In Bangladesh, the incidence of clubfoot is relatively high and access to appropriate treatment is limited. Many affected individuals in Bangladesh are not able to receive the necessary care due to lack of awareness and limited access to healthcare resources

This case study aims to provide an in-depth examination of the current understanding of club foot, including its causes, treatments, and impact on patients and their families, specifically in the context of Bangladesh. In order to gather information for this case study, we visited multiple hospitals and private clinics, but treatment for club foot was not available in all of them.

However we found some facilities, such as Zia Health Complex in Jamalpur, Mymensingh medical college hospital in Mymensingh, Rajshahi medical college hospital in Rajshahi, Fulbaria Upazilla Health Complex in Mymensingh, PG hospital in Dhaka do have the necessary facilities and resources to provide

treatment for clubfoot, Also, there is this NGO named " walk for life"; that is working solely on this disease by providing information and treatment. Additionally, we found that many people in Bangladesh are not familiar with the term "clubfoot" and may not be aware of the condition. To address this gap in knowledge, we also conducted door-to-door interviews with patients who have been diagnosed with club foot to gain insight into their experiences and the challenges they have faced in accessing treatment.

Case Report:

This medical case study report documents the diagnosis and treatment of a patient diagnosed with club foot disease. The patient's medical history, symptoms, and treatment plan were obtained through a series of questions and with the patient's informed consent.

Case-1: Huzaifa, a 3-year-old female, was diagnosed with club foot disease at birth. Both feet were affected. She underwent surgery in her first year and has been using braces and special shoes since. She is currently under regular follow-up care with no significant family history of the condition.

Case-2: Afrin is a 3–5-year-old female patient diagnosed with club foot disease affecting her right leg. Despite being prescribed braces for treatment, she has not yet been able to use them due to financial constraints. There is no known family history of club foot disease. It is noteworthy that Afrin was injured during the delivery process and had delayed milestones such as speaking her first word at 2 years old and walking at 2.5 years old.

Case-3: Mamun is a 46-day-old male patient diagnosed with club foot disease, affecting both feet. Treatment was started 18 days after birth, and the patient underwent tenotomy after 45 days of birth. The patient will begin wearing braces 20 days after the operation. There is no reported family history of club foot disease.

Case-4: Shawon is a 3-year-old male patient diagnosed with club foot disease. He was born with both feet turned inwards and underwent surgery 1.5 years later. He is currently wearing braces as part of his treatment plan. Shawon's family medical history does not indicate any prior cases of club foot disease.

Case-5: Sumaiya is a 2.5-year-old female patient diagnosed with club foot disease. Both of her feet were affected at birth and she was treated with a cast on the 10th day after birth. Later, a tenotomy will be performed. There is no known family history of the condition

Case-6: Yasin, a 2-month-old male, was diagnosed with club foot disease affecting both feet. Treatment for his condition included the use of casts in an attempt to correct his feet without the need for tenotomy. It is noteworthy that Yasin is a twin and his twin is healthy and the patient's father is found to be a smoker. There is no family history of club foot disease reported

Case-7: Yahia, a 34/35-day old male, was diagnosed with club foot disease, in which both feet were affected by birth. Treatment for the condition included casting, which was applied to correct the position of the feet. Yahia had no known family history of club foot disease. The goal of the treatment was to improve the function and appearance of Yahia's feet, and to help him walk more comfortably

Case-8: MD.Amjad Ali Molla is a 55-year-old male who is suffering from club foot disease. Both of his feet are affected by this condition from birth. He has not received any treatment for this condition. His family history reveals that his elder sister and his mother's uncle also suffer from the same type of club foot disease, known as varus. Additionally, it is reported that MD.Amjad Ali Molla is mentally imbalanced, he excretes saliva all the time and also experiences pain while walking.

Case-9: Laboni Akter is an 8-year-old female who has been diagnosed with club foot disease. Her right leg is affected by this condition. The treatment plan for Laboni includes the use of fcrial plaster, special shoes, and exercises. According to her mother, Laboni was normal at birth, but when she first started to walk, this disease was first noticed. The type of club foot that Laboni has is known as equinus. There is no family history of club foot disease in Laboni's family.

Case-10: MD. Al-Amin Hossain is a 22-year-old male who has been diagnosed with club foot disease. Both of his feet are affected by the condition, which he was born with. He is currently undergoing treatment, which includes the use of plaster, special shoes, and exercises. Thankfully, he does not have any family history of the disease. An additional note about MD. Al-Amin Hossain is that he occasionally experiences excessive saliva production, though it is not a constant occurrence. With the proper treatment and therapy, MD. Al-Amin Hossain is expected to make a full recovery and be able to walk normally.

Case-11: The patient, Biplob Ahmmed, is a 1-year-old male who was diagnosed with club foot disease. Both of his feet were affected at birth and he was diagnosed with the condition within 6 months of his birth. The patient was treated with a combination of casting, special shoes, and exercises. The patient's family history did not reveal any cases of club foot disease. The disease type is varus. Additionally, the patient has been observed to excrete more saliva than normal.

Case-12: Moriom Akter, a 2-month-old female, was diagnosed with club foot disease affecting both legs. The patient's condition was characterised as equinus type, which is characterised by a downward pointing foot. The treatment plan included the use of plaster, special shoes, and exercises to help correct the position of the feet. The patient's family history revealed no previous cases of clubfoot

Case-13: Habiba Sultana Jui, a 12-year-old female, was diagnosed with club foot disease affecting both feet from birth. She was treated with casting, shoes and exercise. No family history of the condition was reported. The treatment plan was implemented and regular follow-up appointments were conducted to monitor her progress and adjust the treatment as needed

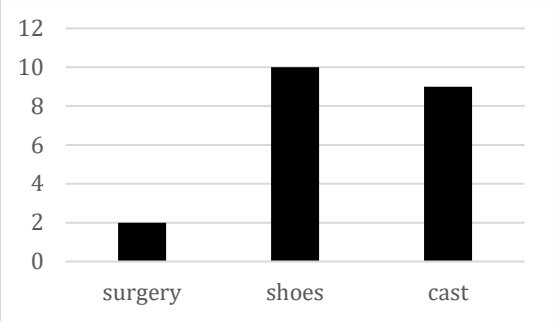
Case-14: Jannatul Ferdows is a 2-year-old female who was diagnosed with club foot in both of her feet at birth. The Ponseti method, which includes serial casting and bracing, was used to treat her condition. her parents reported that they noticed improvement in her feet within the first few weeks of treatment.

Case-15: Rakibul Hasan is a 6-month-old male who was diagnosed with club foot in his left foot at birth. His mother has a history of club foot, so the family was aware of the condition. The Ponseti method, which includes serial casting and bracing, was used to treat his condition.

Case-16: Ahmed Farukh is a 4-year-old male who was diagnosed with club foot in both of his feet at birth. The Tendo-Achilles lengthening and heel cord release surgery, was used to treat his condition. Ahmed's treatment process was done in several stages, firstly he underwent the Tendo-Achilles lengthening and heel cord release surgery, then he was put on bracing

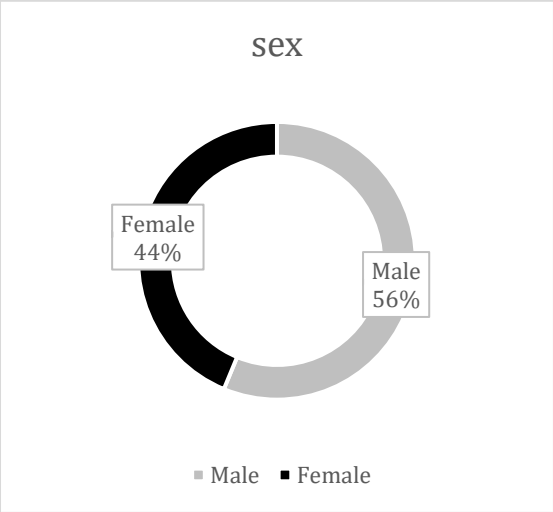
These case studies demonstrate the varied presentations and outcomes of club foot in patients from different region of bangladesh. Early diagnosis and treatment with the Ponseti method, which is considered the gold standard, can result in successful correction of the condition. However, some patients required additional interventions or ongoing management to achieve optimal outcomes.

The chart below shows, what treatment method the patients took (some went through several)



From the graph we can see that, surgery has been done for treatment for only two case (case-3 and) and in the rest of the cases Ponseti method is used.

From our findings we found 7 female and 9 males affected by this condition.

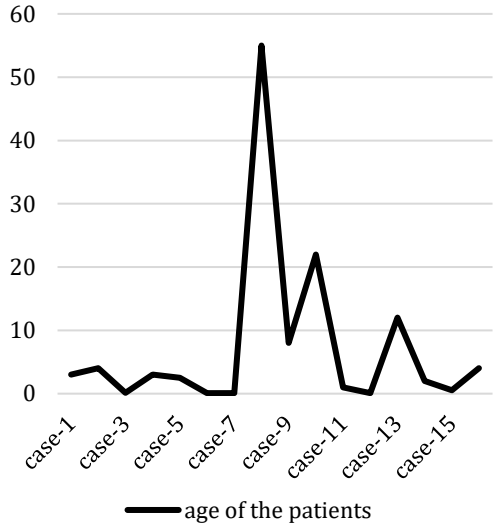


It is important to note that most patients were from low-income families. This highlights the potential impact of socioeconomic factors on access to and outcomes of treatment for clubfoot. These families may have limited access to medical care and resources, which can delay diagnosis and treatment, as well as affect adherence to treatment plans. It's also important to mention that lower class families may face financial difficulties to afford the costs of treatment and rehabilitation. In addition to the socioeconomic factors, it was also found that some patients had mothers who did not receive proper prenatal care during their pregnancy. This lack of care during pregnancy may have contributed to the development of club foot in these children.

The exact cause is usually not identified. Both genetic and environmental factors are believed to be involved. We only found two case; the case-8 of MD.Amjad Ali Molla,and case-15 of Rakibul Hasan where the disease seemed to be inherited,

There appears to be hereditary component for this birth defect given that the risk of developing congenital clubfoot is 25% when a first-degree relative is affected.

It should be noted that age of patients in this case study varied a lot, they were between the ages of 35 days and 55 years. This age range is typically when clubfoot is most commonly diagnosed.



We visited a number of hospitals across Bangladesh to investigate the availability of treatment for clubfoot, and we found that not all of the hospitals had the necessary facilities to treat this condition. This is concerning, as according to the National Strategy and Action Plan for Clubfoot Care in Bangladesh, the country has a network of Ponseti Clubfoot clinics at 44 medical college and district hospitals, managed by national and international non-government organizations. However, only 3 of these clinics are specialized hospitals, and 24 are secondary level hospitals. This is a very low number of facilities to provide treatment to the patients who are suffering from clubfoot.

Furthermore, we also found that the cost of treatment for clubfoot is a significant barrier for many patients. The cost to make each pair of clubfoot treatment shoes varies from 3 to 5 dollars, and each pair is sold between 5 to 10 dollars. This is high for patients from low-income families, making it difficult for them to afford the treatment.

Discussion: the study highlights the importance of providing access to adequate prenatal care for all women, particularly those from low-income families, to minimise the risk of birth defects and other complications in their children. Prenatal care should include proper nutrition, exercise, and awareness of risk factors for birth defects, as well as regular check-ups with a healthcare provider to monitor the health of both the mother and the developing foetus. It is important for healthcare providers to be aware of the signs and symptoms of club foot, particularly in young children, to ensure prompt diagnosis and treatment. Early intervention can greatly improve the outcome of the treatment and prevent long-term complications. In addition to that, the need for increased availability and accessibility of treatment for clubfoot in Bangladesh. The low number of facilities and the high cost of treatment are major barriers for patients, and these issues need to be addressed in order to provide better care for those suffering from clubfoot. The National Strategy and Action Plan for Clubfoot Care in Bangladesh should be implemented effectively to improve the clubfoot situation in Bangladesh.

Conclusion:

this medical case study report has documented the diagnosis and treatment of 16 patients with club foot disease. The patients were diagnosed at different stages of their lives, ranging from birth to adulthood. The treatment plans for each patient varied, including surgery, casting, and bracing. Despite the different treatment plans, patients' medical histories, symptoms, and treatment plans were obtained with their informed consent. Additionally, it is observed that some of the patients have a family history of club foot disease, while most of them have no known family history. the study conducted by visiting a number of hospitals across Bangladesh highlights the lack of availability of treatment for clubfoot. Despite the existence of a network of Ponseti Clubfoot clinics, only a small number of these clinics are specialized hospitals, and the majority of them are secondary level hospitals. This is a concerning issue, as it means that not all patients who are suffering from clubfoot have access to the necessary facilities and treatment options. The National Strategy and Action Plan for Clubfoot Care in Bangladesh aims to improve this situation, but as of now, the number of facilities providing treatment for clubfoot remains low. It is crucial that more specialized hospitals and facilities be established to provide better treatment options for patients with clubfoot and to ensure that all patients have access to the care they need.

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